

Lab Goal : This lab was designed to teach you more about graphs, mazes, and recursion.

Lab Description : After successfully retrieving your soul, you decide to leave quickly before you can get into any more trouble. Unfortunately, one of the tournament organizers seems to have noticed what you did, and are now after you. If you are able to get home before they get there, though, you will be safe. You observe their speed to be a constant, and calculate are therefore able to calculate how quickly they will arrive at your house since you know how far your house is and it's a single road path from the convention center to your home. Unfortunately, you can't take the main road because there are tournament organizers watching it. Instead, you have a number of alternate routes you can take, which do not directly connect your house and the convention center, but may actually be faster despite your detours through other destinations. Given a graph of these alternate paths, with edge lengths being the distance from one destination to another, and the distance the organizers must travel to get to your house, can you find out if you will make it home before they do?

Input : The first line will contain a single integer n that indicates the number of data sets that follow. Each data set will start with an integer l , the distance of your house from the convention center via the main road, and x , the length and width of the adjacency matrix to follow that contains the graph of possible alternate routes. The next x lines will contain this matrix. The convention center will always be the first node, and your house will always be the last node. The main road is not included in the adjacency matrix. In addition, the cost to remain on the same node will always be 0 because it does not take any time to stay where you are.

Output : If you are able to make it from the convention center (first node) to your house (last node) by traveling a shorter distance than the tournament organizer does using the main road, output "Safe at last." If you take an equal or longer distance than the tournament organizer, output "This is the end of me!"

Example Input File

```
6
3 5
1 1 1 0 0
1 1 0 0 0
1 0 1 1 1
0 0 1 1 0
0 0 1 0 1
4 5
1 1 0 0 0
1 1 1 0 0
0 1 1 1 0
0 0 1 1 1
0 0 0 1 1
5 10
1 1 0 0 0 0 0 0 0 0
1 1 1 0 0 0 0 0 0 0
0 1 1 1 0 0 0 0 0 0
0 0 1 1 1 0 0 0 0 0
0 0 0 1 1 1 0 0 0 0
```

Files Needed ::

You must create the java file.

The data file is provided.
run.dat

```

0 0 0 0 1 1 1 0 0 0
0 0 0 0 0 1 1 1 0 0
0 0 0 0 0 0 1 1 1 0
0 0 0 0 0 0 0 1 1 1
0 0 0 0 0 0 0 0 1 1
5 10
1 1 0 0 0 0 0 0 0 1
1 1 1 0 0 0 0 0 0 0
0 1 1 1 0 0 0 0 0 0
0 0 1 1 1 0 0 0 0 0
0 0 0 1 1 1 0 0 0 0
0 0 0 0 1 1 1 0 0 0
0 0 0 0 0 1 1 1 0 0
0 0 0 0 0 0 1 1 1 0
0 0 0 0 0 0 0 1 1 1
1 0 0 0 0 0 0 0 1 1
1 1
1
0 1
1

```

Example Output to Screen

```

Safe at last.
This is the end of me!
This is the end of me!
Safe at last.
Safe at last.
This is the end of me!

```